

National Higher School of Mathematics

Preparatory Cycle
First Year

Academic Year : 2024/2025
Course : Algebra 1

Worksheet N°3

Binary relations

The starred exercises are to be done in tutorial sessions

Exercise 1.

Let S be a nonempty set, A a subset of S and $\mathcal{R}$ a binary relation defined on $\mathcal{P}(E)$ by

$$\forall X, Y \in \mathcal{P}(E) : X\mathcal{R}Y \iff X \cap A = Y \cap A.$$

1. Prove that $\mathcal{R}$ is an equivalence relation.
2. Determine $\overline{\emptyset}$, $\overline{S}$.

Exercise 2. ★

Let $\mathcal{R}$ be a binary relation defined over $\mathbb{R}$ as follows

$$\forall x, y \in \mathbb{R}, x\mathcal{R}y \iff x - y \in \mathbb{Z}.$$

1. Verify that $\mathcal{R}$ is an equivalence relation.
2. Determine $\overline{x}$ for any real number x .
3. Show that $\mathbb{R}/\mathcal{R} = \{\overline{x} \mid x \in [0, 1[\}$.

Exercise 3. ★

Let $\mathcal{R}$ be the binary relation defined on $\mathbb{R}$ by $x\mathcal{R}y$ if and only if $\cos^2 x + \sin^2 y = 1$.

1. Show that $\mathcal{R}$ is an equivalence relation. Is $\mathcal{R}$ a partial order relation?
2. Determine the equivalence classes $\text{cl}\left(\frac{\pi}{3}\right)$ and $\text{cl}\left(\frac{\pi}{2}\right)$.
3. Determine $\mathbb{R}/\mathcal{R}$.

Exercise 4. ★

Let X and Y be two sets, and let $f : X \rightarrow Y$ be a map. Define a relation $\mathcal{R}$ on X by

$$\forall x, y \in X, x\mathcal{R}y \iff f(x) = f(y).$$

1. Prove that $\mathcal{R}$ is an equivalence relation, and describe the equivalence class $\overline{x}$ for any element $x \in X$.
2. Now consider the map

$$\begin{aligned} \overline{f}: X/\mathcal{R} &\longrightarrow Y \\ \overline{x} &\longmapsto f(x) \end{aligned}.$$

- (a) Show that $\overline{f}$ is well-defined.
- (b) Prove that $\overline{f}$ is injective.
- (c) What can you conclude about the quotient set?

Exercise 5. ★

We define a binary relation $\mathcal{R}$ on $\mathbb{R}^2$ by

$$(x, y)\mathcal{R}(x', y') \iff x + y = x' + y'.$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Describe the equivalence class of $(0,0)$, and then describe all equivalence classes.
3. Consider the map $\phi : \mathbb{R}^2/\mathcal{R} \rightarrow \mathbb{R}$ defined by $\phi(\overline{(x,y)}) = x + y$, where $\overline{(x,y)}$ represents the equivalence class of (x,y) . Verify that ϕ is well-defined and that ϕ is bijective.

Exercise 6 (Real projective line). $\star$

Let $\mathcal{R}$ be the binary relation on the set $\mathbb{R}^2 \setminus \{(0,0)\}$ defined as follows :

$$\forall M, M' \in \mathbb{R}^2 \setminus \{(0,0)\}, M \mathcal{R} M' \iff \exists \lambda \in \mathbb{R}^*, \overrightarrow{OM} = \lambda \overrightarrow{OM'}.$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Describe the equivalence classes and the quotient set $\mathbb{RP}^1 = (\mathbb{R}^2 \setminus \{(0,0)\})/\mathcal{R}$.
3. Show that there exists a bijection between $\mathbb{RP}^1$ and the circle with center $(1,0)$ and radius 1.

Exercise 7. $\star$

We define on $\mathbb{N}$ the relation $\mathcal{R}$ by

$$\forall p, q \in \mathbb{N}, p \mathcal{R} q \iff \exists n \in \mathbb{N}, q = p^n.$$

1. Show that $\mathcal{R}$ is an order relation. Is it a total order?
2. Is the subset $\{2,3\}$ upper bounded.
3. Let $A = \{2,4,16\}$. Determine the maximal and minimal elements of A .

Exercise 8 (Lexicographical order). $\star$

We define on $\mathbb{R}^2$ the relation $\mathcal{R}$ by:

$$\forall (x,y), (x',y') \in \mathbb{R}^2, (x,y) \mathcal{R} (x',y') \iff ((x < x') \text{ or } (x = x' \text{ and } y \leq y')),$$

where $\leq$ is the usual order relation on $\mathbb{R}$.

1. Show that $\mathcal{R}$ is an order relation¹. Is it a total order?
2. Determine the set of upper bounds and lower bounds of $\{(a,b)\}$ and represent them in $\mathbb{R}^2$.
3. Let $X = \{(a,b), (c,d)\}$. Determine $\sup X$ and $\inf X$.
4. Define the lexicographical order of $\mathbb{R}^3$, then generalize to $\mathbb{R}^n$.

Exercise 9.

1. Let $\mathcal{R}$ be an equivalence relation on a set S . Show that if C_x is the equivalence class of $x \in S$, then we have

$$\forall x, y \in S, x \neq y \implies (C_x = C_y \text{ or } C_x \cap C_y = \emptyset).$$

(ie. the equivalence classes form a partition of the set S)

2. Let S be a set. Show that the inclusion " $\subset$ " is an order relation on the set $\mathcal{P}(S)$. Is $(\mathcal{P}(S), \subset)$ a totally ordered set?
3. Let S be a totally ordered set. We say that S is well-ordered if $\forall A \subset S, A \neq \emptyset$, there is $m \in A$, such that: $\forall x \in A, m \leq x$.
 - (a) Show that the set $\mathbb{N}$ is well-ordered.
 - (b) Show that the sets $\mathbb{Q}$ and $\mathbb{R}$ are not well-ordered.

1. $\mathcal{R}$ is called lexicographical order

Exercise 10 (Combination of relations).

Let S be a set.

1. Let $\leq$ be an order relation on S . Let $\prec$ be the binary relation on $\mathcal{P}(S) \setminus \{\emptyset\}$ defined as follows:

$$\forall A, B \in \mathcal{P}(S) \setminus \{\emptyset\}, A \prec B \iff \forall x \in A, \forall y \in B, x \leq y \text{ or } A = B.$$

Show that $\prec$ is an order relation.

2. Let $\mathcal{R}_1$ and $\mathcal{R}_2$ be two relations on S . We define the two relations $\mathcal{T}_1$ and $\mathcal{T}_2$ on S by the following expressions:

$$x\mathcal{T}_1y \iff x\mathcal{R}_1y \wedge x\mathcal{R}_2y$$

$$x\mathcal{T}_2y \iff x\mathcal{R}_1y \vee x\mathcal{R}_2y.$$

$\mathcal{T}_1$ is called the intersection of $\mathcal{R}_1$ and $\mathcal{R}_2$. Similarly, $\mathcal{T}_2$ is called the union of $\mathcal{R}_1$ and $\mathcal{R}_2$.

- (a) Show that the intersection of two equivalence relations is an equivalence relation. What can we say about their union?
 - (b) Show that the intersection of two partial order relations is a partial order relation. Give an example where the union of two partial order relations is not a partial order relation.
3. Let $\mathcal{R}$ be a reflexive and transitive relation on S . We define the relations $\mathcal{T}'_1$ and $\mathcal{T}'_2$ as follows:

$$x\mathcal{T}'_1y \iff x\mathcal{R}y \wedge y\mathcal{R}x$$

$$x\mathcal{T}'_2y \iff x\mathcal{R}y \vee y\mathcal{R}x.$$

- (a) Show that $\mathcal{T}'_1$ is an equivalence relation. Can we say the same thing about $\mathcal{T}'_2$?
- (b) Show that $\mathcal{R}$ allows us to define an order on $S/\mathcal{T}'_1$.

Exercise 11 (Trellis). ★

Let S be a set and $\leq$ a reflexive and antisymmetric relation over S such that any pair of elements $(a, b) \in S^2$ has a supremum denoted $(a \wedge b)$ and an infimum denoted $(a \vee b)$. A set with such a property is called trellis, and is denoted $(S, \leq, \wedge, \vee)$.

Show that the following statements are equivalent

$(S, \leq, \wedge, \vee)$ is a trellis.

The operations $\wedge$ and $\vee$ are associative.

One of the two operations $\wedge, \vee$ is associative.